

A Personalized Real-Time Vocabulary Coaching Assistant for Second Language English Writers

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Abstract

The writing assistants powered by artificial intelligence are currently commonly employed in the stylistic refining of the text and grammatical error correction. However, most of the existing tools are focusing on automated rewrites or superficial corrections, which do not offer much learning to users who are learning English as a second language (ESL). This project is aimed at developing a real-time, personalized vocabulary coaching assistant, which can only be used as a text-proofreading aid, but not as a text-generating system. In live writing, the system identifies non-optimal lexical options, provides contextually applicable alternatives and provides brief explanations and examples of each proposal. The suggested assistant aims at reducing the distance between correction of automated writing and effective acquisition of the second language through emphasis on transparency, user learning and vocabulary development.

1 Introduction

The recent breakthroughs in natural language processing have resulted in the wide usage of AI-based writing assistants across scholars, work, and personal settings. Grammar checkers, paraphrasing systems and other tools can quickly make a person more fluent and correct, but they are usually black-box systems that do not encourage a deeper understanding of the language. This becomes a serious constraint to those studying English as a second language: errors are being rectified, and vocabulary development and in-depth understanding of the language is often missed.

The empirical research in second language acquisition focuses on the idea that vocabulary acquisition is a key factor that defines the writing competence, reading comprehension, and long-term language recovery. This is notwithstanding that the majority of commercial writing aids concentrate on restructuring sentences, modulating tone, or fully writing a piece of writing instead of focused training in vocabulary. Additionally, text generation systems that create text directly pose a threat to agency on the part of learners and are likely to foster passive dependency on technologies instead of active capacity building.

This project would fill these gaps by suggesting a vocabulary coaching helper which is real time and only interferes on the level of lexical choice. The system identifies any generic or imprecise use of words by the user as he or she writes, and suggest better quality alternatives that seem to fit in the context of the

sentence and the level of proficiency the user has. Most importantly, there is a concise explanation on why the alternative is better and a sample sentence on how the alternative would be used. This design is such that corrections can be read, learnable, and taken up at once.

The proposed assistant, unlike the current tools, will not produce original content, paraphrase whole sentences, or distort the intention of the author. Rather, it plays the role of a smart tutor that is part of the writing process. This system will assist in vocabulary learning with the integration of real-time feedback, personalization, and clear pedagogical explanations without losing user control of the writing process.

2 Problem Statement

The principle problem that the given project is trying to address is that existing AI writing aids in second language English learners do not allow providing real-time, pedagogically-based feedback on vocabulary. The current tooling either focuses on grammatical precision or provides the general suggestions without clarifying the linguistic reasoning behind the change in vocabulary. Consequently, users can be short-term better writers but fail to increase their vocabulary and become better writers independently.

The hypothesis of this work is that a personalized, explanation-driven vocabulary coaching system can improve lexical sophistication and long-term learning outcomes for ESL writers without relying on automated text generation.

3 Motivation

The knowledge of vocabulary is the fundamental aspect of good written communication, especially when it comes to non-native speakers of English in a school or workplace. The problem with learners is that learners do not have ideas, but are unable to find the right words with the right context to convey these ideas. The solutions to this challenge are of practical value as they can lead to better academic results, greater confidence in writing, and minimized reliance on automated rewriting programs.

In a broader scope, the study will help in building human-oriented artificial intelligence systems that can focus on learning and transparency rather than automation. This project meets the criteria of ethical AI and educational best practices by creating an assistant that has the ability to explain its actions, as well as, uphold the agency of its users. Practical relevance of the proposed approach to the language education platforms, productivity tools in the workplace and writing technologies oriented on accessibility are also suggested.

References

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